

Welcome to 151B: Deep Learning

Instructor: Annapurna Vadaparty

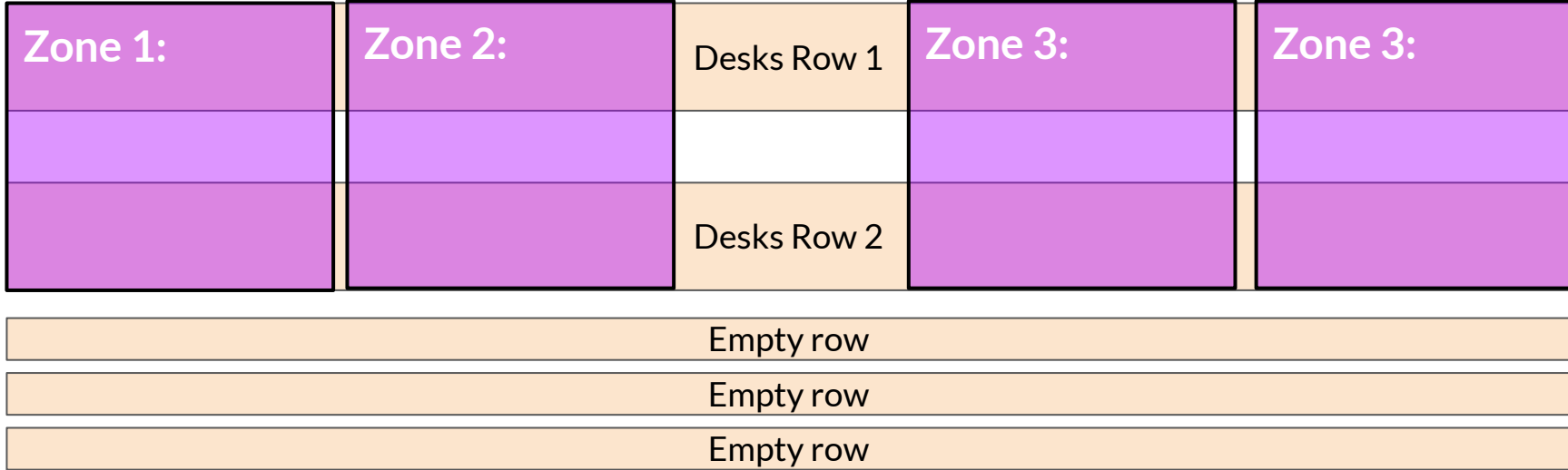
TA: Ben Bao

Tutors: Alexander Radulescu, Trevor Duong

Seating

I fractured my foot which is why we're on Zoom today.
Bear with me :)

Podium



1. Introduce yourself to the other people in your zone
2. How is A1 going so far? Any points where your stuck?
Can other people in your zone help?

Welcome!

- A few new students–would be great if everyone could help them get settled & explain things they missed!

I have fewer spoons for awhile than I was expecting to...

Spoon theory

🌐 12 languages ▾

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From Wikipedia, the free encyclopedia

For the socioeconomic idea, see [spoon class theory](#).

Spoon theory is a concept used to represent how individuals manage limited energy. It is a [metaphor](#) describing the amount of [physical](#) or [mental energy](#) that a person has available for daily activities and tasks, and how it can become limited. The term was coined in a 2003 essay by American writer Christine Miserandino. In the essay, Miserandino describes her experience with [chronic illness](#), using a handful of [spoons](#) as a metaphor for units of energy available to perform everyday actions. The metaphor has since been used to describe a wide range of [disabilities](#), [mental health](#) issues, forms of [marginalization](#), and other factors that might place unseen burdens on individuals. In conjunction with spoon theory, members of the chronic illness community have developed several related theories to describe other aspects of energy management.



[Spoons](#) are used as a metaphor for energy rationing. 

Origin [\[edit \]](#)

In her 2003 essay "The Spoon Theory", American writer Christine Miserandino writes about a time she told a friend about her experience with [lupus](#). As they were at a restaurant, Miserandino took a number of spoons from the tables and handed them to her friend. Miserandino said that while most people begin each day with an "unlimited amount of possibilities", people with chronic illness have to plan their actions in order to conserve their energy. The number of spoons represented how much energy was available to spend throughout the day. Miserandino asked her friend to list the different tasks and errands that she might perform on a typical day, with Miserandino interrupting and taking away a spoon for each activity – such as washing her hair, or standing on a train – that required some effort or would otherwise be affected by [lupus](#).^[1]

The plan going forward

- Remote for today
- I should be able to be back in person tomorrow
- I will only be on campus in general Mon-Weds–your wonderful tutors and TA will be taking over oral assessments on Thursdays

Things that would really help me out

- Please stick to contacting me through Piazza except for personal issues
 - Don't email about missed deadlines, etc.
 - I have a lot of emails and I may miss yours!
- Work with Ben, Alex, and Trevor to come up with plans that work for everyone's schedules
 - To the extent possible/what you're comfortable with, share what constraints you have. We will try our best to figure out plans that work for everyone.

A1 Submissions

- A Google Drive folder link will be posted shortly for you to upload your A1 submission
 - (To my knowledge) Deep Learning/ML Classes are not usually autograded, so thank you for your patience as we get this all set up! It seems like the discussions we have in class are a worthwhile tradeoff for staff time!
- A1 Algorithms & Architecture question submissions will be on PL soon (later today)

What did we learn from the practice tests?

CBTF/TTC	Oral Assessments
Come on time. Proctors are not guaranteed to wait for you, and are unlikely to wait if you're the only person signed up for your slot. This happened during one of the practice tests!	Exams are in the CSE basement—mentioned on the website and calendar invites (room change was which basement room).

Oral Assessment

- They're only for the programming portion (the architectures & algorithms portions will be assessed in CBTF; oral assessments are how we know that you've understood the programming portion)
- Think through (ideally aloud), take your time
- High pass isn't judging how quickly you solve it. We do want you to get it right on the first try, though (this helps with fairness—make it easy for us to say that you had a high pass!)
- Work through it aloud and make mistakes along the way before you commit to answer that you tell us.
 - **We won't grade anything you say until you say “My answer is...”**
 - **We'll only give hints/prompts if you say yes to “Do you need help?”**

How was the first homework and oral assessment?

Some things I learned from your reflections (LPRR):

- I'm impressed how many of you really try to do the problems yourself before looking at answers!
- Past classes help a lot–this is great. If you're open to sharing specific links from past classes that your classmates can use, even better!
- Would be great to post concrete things classmates could use:
 - Actual **links** to sites/resources you used
 - GenAI **prompts** you used (e.g. to encourage it to hint rather than give you answers)
 - ...anything else?

Until this Friday: You can update/post your A0 LPRR

Standards: Learning Process & Resource Reflections

LPRR standards

Grade	Standard
HP	At least 300 words, all questions completed
P	At least 200 words, all questions completed
LP	Some questions not completed
NP	Above criteria not met

Make sure your name is visible to instructors.

In general, make sure to check the deadlines & late deadlines.

A

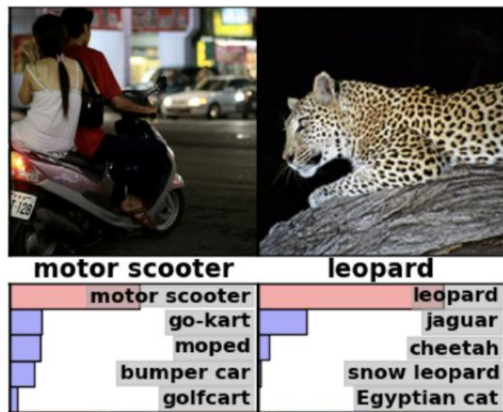
- Above 80% on both exams
- HP on at least 3 of 4 assignments, P on the remaining one
- **HP on all Learning Process & Resource Reflections**
- Completed 80% of pre-lecture homeworks for completeness, before class
- HP on 80% of pre-lecture homeworks for correctness, P on the rest
- HP on the project

Other deadline/logistics questions?

- Project proposals ready?
- Anything else?

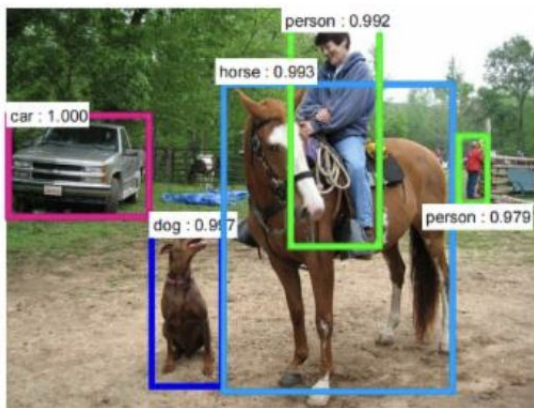
Images: What kinds of problems would we want to solve?

object recognition



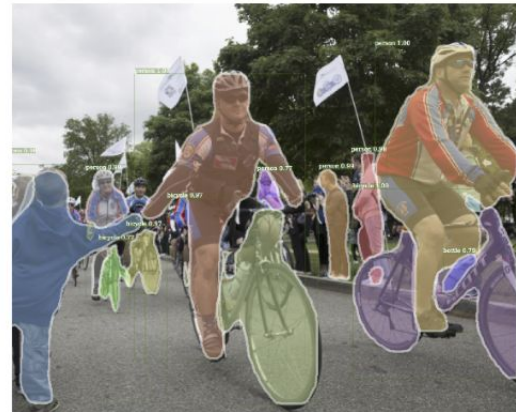
Krizhevsky et al., 2012

object detection



Ren et al., 2016

object segmentation



He et al., 2017

What are some things we know about images?

we can exploit known structure in spatial and sequential data
to impose priors (i.e. inductive biases) on models

inductive: inferring general laws from examples

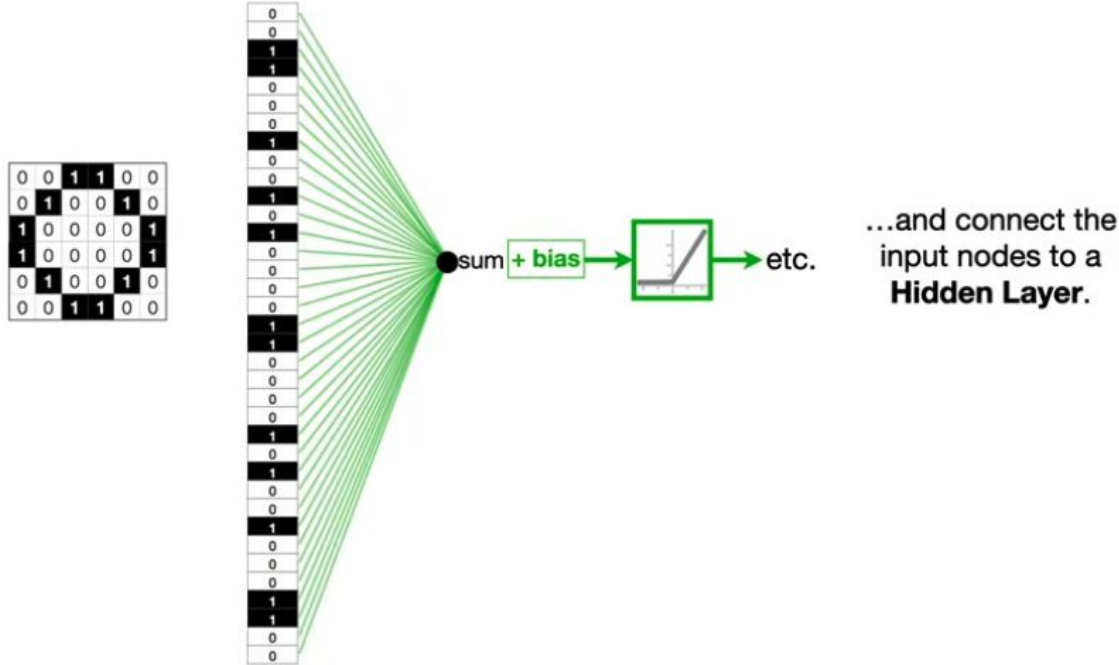
What are some known structural properties of images generally?

Properties of the visual world

Four important properties:

- Nearby pixels correlate the most with nearby pixels – not pixels far away:
locality
- On the other hand, the statistics of pixels are relatively uniform across the image: **stationary statistics**
- Identity of an object (usually) doesn't depend on its location in the image:
translation invariance
- Objects are made of parts: **compositionality**

Convolutions: Why do we need them?

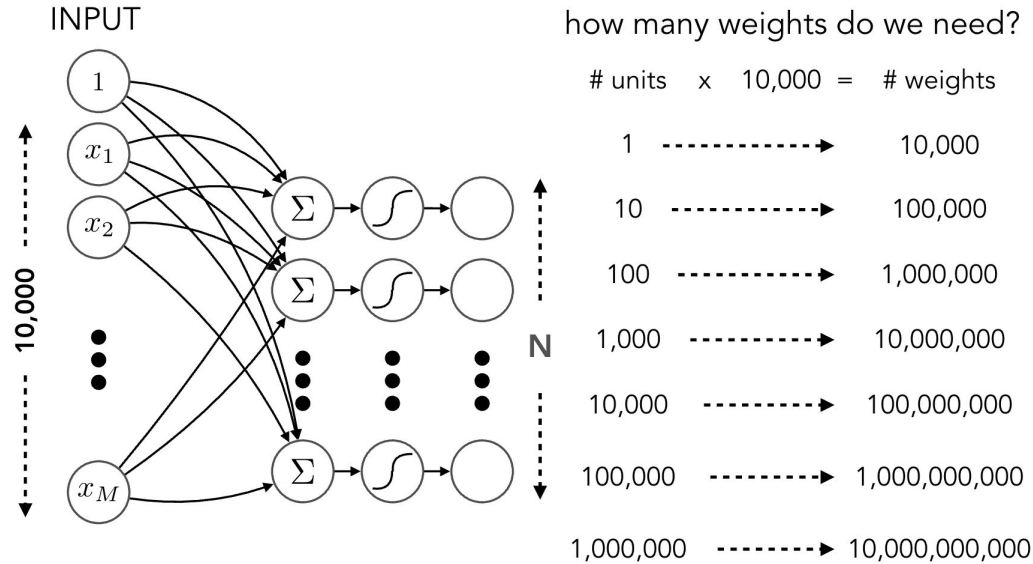


What's happening in this picture?

We have huge NNs all the time...why is this bad?

What does an input feature in a MLP usually represent?
Does a pixel carry that same amount of information on its own?

Infeasible to do this for even a few basic patterns



*if we want to recognize even a few basic patterns at each location,
the number of parameters will explode!*

How do CNNs help instead of doing the process on the previous slide?

- What do you think?

How do CNNs help instead of doing the process on the previous slide?

- Fewer input nodes
- Recognizes small shifts
- Use correlations

How does the filter help us achieve those?



Input Image

0	0	1	1	0	0
0	1	0	0	1	0
1	0	0	0	0	1
1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0

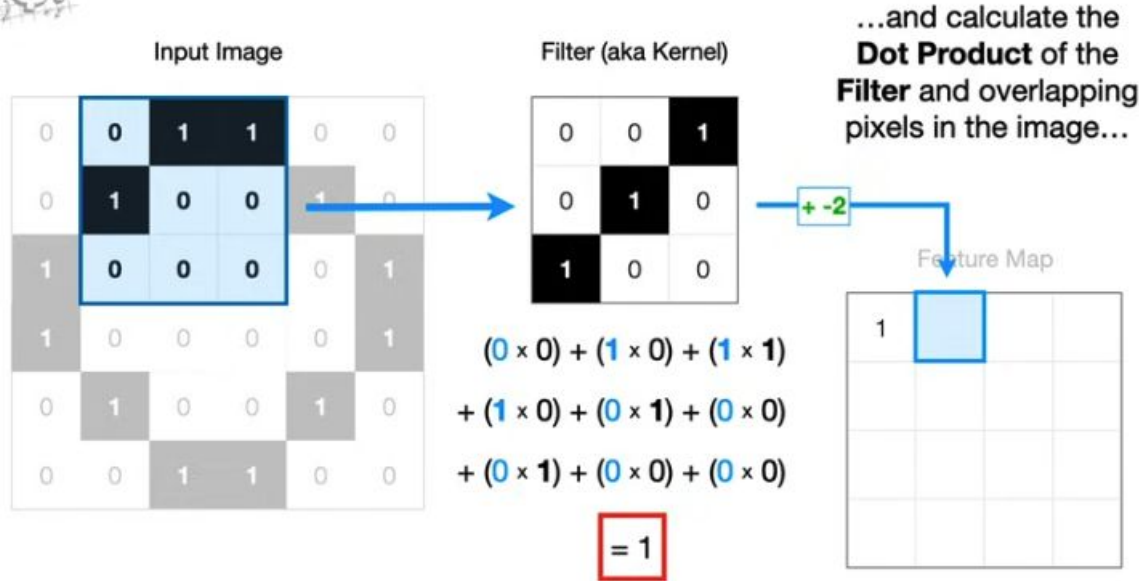
Filter (aka Kernel)
3 pixels wide

0	0	1
0	1	0
1	0	0

3 pixels tall

In **Convolutional Neural Networks**, a **filter** is just a smaller square that is commonly **3 pixels by 3 pixels**...

1. Fewer input nodes

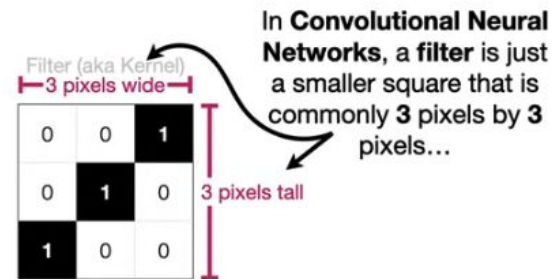
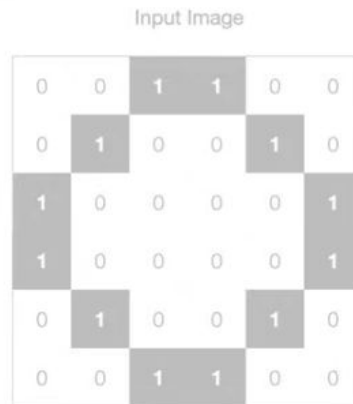


How many pixels in the original image?

How many datapoints in the feature map?

2. Recognizing Small Shifts

- This one is hard to see with just one pass of the kernel
- But we can start to see it even there—what does/doesn't change if we move the filter?



3. Uses correlations

- What does the filter end up looking like? Why did backprop get it from here to here?

1	0	1
0	0	1
1	1	0

(Before Training)

0	0	1
0	1	0
1	0	0

Convolution Arithmetic

filter weights: $\begin{pmatrix} 0 & 1 & 2 \\ 2 & 2 & 0 \\ 0 & 1 & 2 \end{pmatrix}$

3_0	3_1	2_2	1	0
0_2	0_2	1_0	3	1
3_0	1_1	2_2	2	3
2	0	0	2	2
2	0	0	0	1

$$(f * g)(x, y) = \sum_{i=1}^I \sum_{j=1}^J f(i, j)g(x + i - 1, y + j - 1)$$

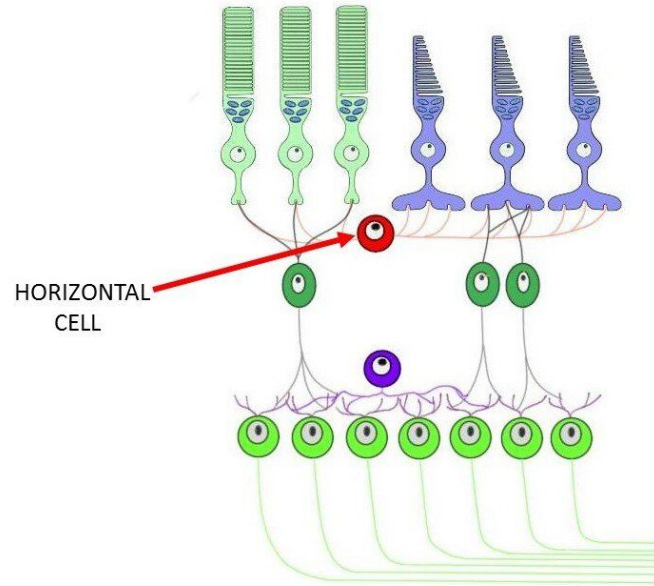
take inner (dot) product of filter and each input location
measures degree of filter feature at input location

→ *feature map*

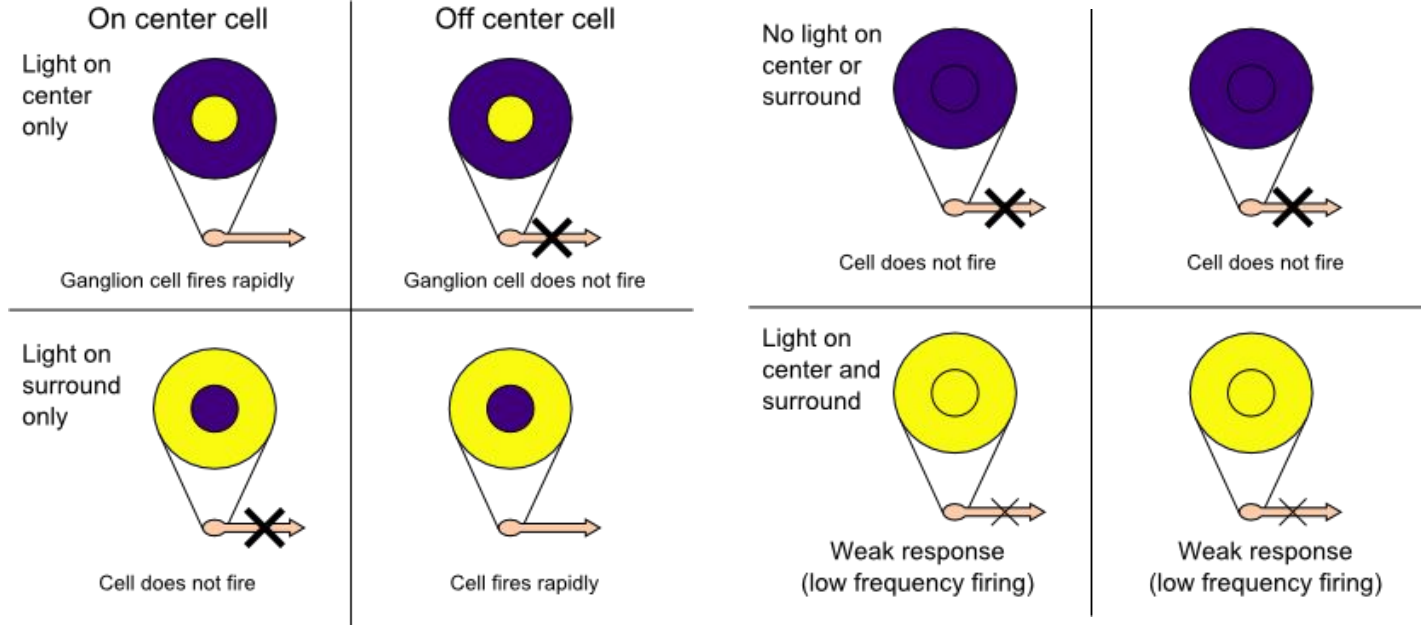
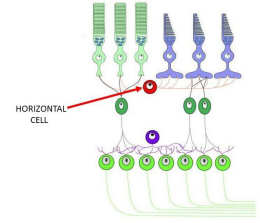
Break!

See you in 5 mins.

How do eyes do edge detection?



How do eyes do edge detection?



What do these cells remind you of?

What is a receptive field?

In neuroscience a receptive field of a neuron refers to both:

- The *size* of the visual world that most stimulates the neuron
- The *visual features* in that part of the visual world that most stimulates that neuron (e.g. an “edge”)

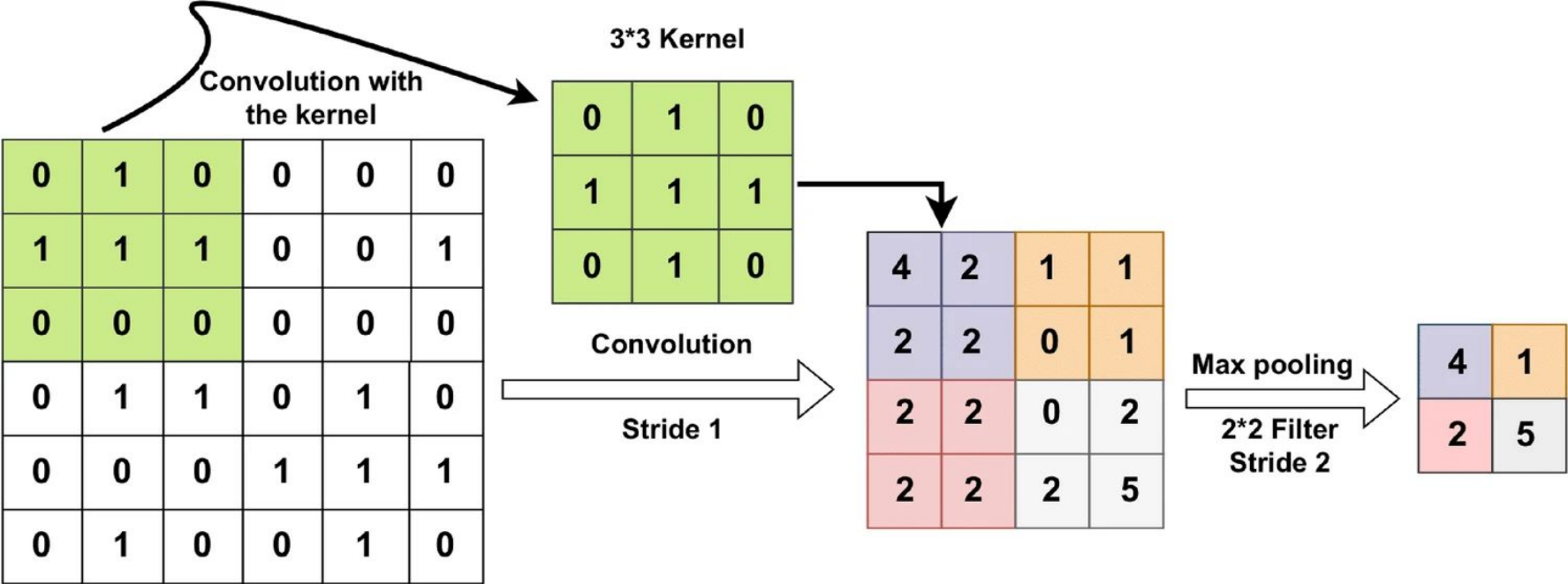
We will usually refer to the first meaning as receptive field; the second we'll just call the features, or the kernel, or the filter. They all mean the same thing.

How do CNNs leverage properties of images?

CNNs began with LeCun et al. in 1989—heavily biologically inspired!

1. Small, local receptive fields and learned features (kernels): **locality**
2. These are replicated across the image: **stationary statistics**
 - Perhaps a better intuition: If a feature is useful in one place, it's useful in others

Pooling



How do CNNs leverage properties of images?

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1. Small, local receptive fields and learned features (kernels): **locality**
2. These are replicated across the image: **stationary statistics**
 - Perhaps a better intuition: If a feature is useful in one place, it's useful in others
3. Spatial pooling: **translation invariance**
4. Receptive fields get larger deeper in the net: Objects are made of parts - **compositionality**

Questions?

End of where we reached in class

Follow-Ups/Corrections

- None for today!